

Welcome to Section One of Module Two

In this section you will learn about

- * the various measures of central tendency
- * the main measures of variability
- * validity and reliability

Means, Modes, and Other Measures of Central Tendency

After collecting a set of numerical data (or data that can be converted to numerical values), researchers often calculate averages to help them interpret information and predict events. Basic statistical concepts such as the mean, mode, and median are the most common measures of **central tendency** used by researchers. Central tendency refers to how clustered a group of numbers is. Mathematically, central tendency refers to *the average* in a set of data.

In addition to researchers, many other individuals use measures of central tendency in various aspects of their jobs. For example, teachers calculate averages (report card marks), meteorologists make weather comparisons (current temperature to average value), and coaches determine who to place on the opening roster (batting averages).

One of the top three reasons people watch the news is the weather report. To illustrate the main measures of central tendency, consider an analysis of the weather for 11 consecutive days in May, specifically, the temperature values for a small town in Northern Alberta in the table below.

Sun	Mon	Tue	Wed	Thur	Fri	Sat	Sun	Mon	Tue	Wed
										
11 °C	16 °C	15 °C	14 °C	15 °C	19 °C	12 °C	15 °C	13 °C	14 °C	10 °C

central tendency: a measure of mathematical averages such as mean, mode, and median

Mode, Median, and Mean

The **mode** is the number that occurs most often in a set of data. To determine the mode, you should first organize the numbers in order of size. Using our temperature values from the previous page, the new order would be as follows:

10 °C	11 °C	12 °C	13 °C	14 °C	14 °C	15 °C	15 °C	15 °C	16 °C	19 °C
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The **mode** is

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This number occurs three times.

The median is the number in the middle of the set of data after you arrange the numbers in order of size. In many cases, you will have the same number of values on the left and right of the median value. If, however, there is an even number of data points, you will have two numbers as your median and the values must be averaged. In our case, we have 11 data points. The median is data point six.

The **median** is

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There are five data points to the left and five data points to the right of the median.

The mean is the value obtained when you add all the values in a data set and then divide the sum (total) by the number of values in the data set. In our case, there are 11 values.

When added, the 11 values equal

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You then divide the above total by 11 to get a **mean** of

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mode: the number that occurs most often in a set of data

median: the number in the middle of an ordered set of data

mean: the value obtained by dividing the sum of all values by the total number of values

Required Practice: Mean, Mode, Median

To answer some of the questions in Assignment Three, you must access the LearnAlberta.ca website (<http://learnalberta.ca/>) and complete several exercises. Content on LearnAlberta.ca, while created for all Albertans, is accessible only with a username and password. Passwords and usernames change every year. If you have forgotten the password for this year, please contact your teacher, marker, or school. *

*Your username and password may automatically be entered for you, once you enter the link.

Please read the following instructions in their entirety B E F O R E entering the LearnAlberta.ca website and completing the exercises.

Step One:

Enter the LearnAlberta.ca website and chose your preferred language. IF you need to Login as an Authorized User, you will need to enter a valid Username and Password. The Username is LA48 The Password is 1958.	Choose your preferred language first  Choose your language Choisir la langue English Français Next, sign in Sign in * Username: Your Email or Jurisdiction Username * Password: Sign in
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Step Two:

In the *Find Resources* section there are three drop-down menus.

Please enter "central tendencies" in the Enter Keyword box.

Click "Search"

Find Resources

[More Search Options](#)

New! Find resources by [Program of Study!](#)

Step Three:

Scroll to the right-hand side of the page and look for *resources matching your criteria*.

You may find a few matches. Click on the [Central Tendencies](#) match-link.

Resources matching your criteria:
central tendencies

5

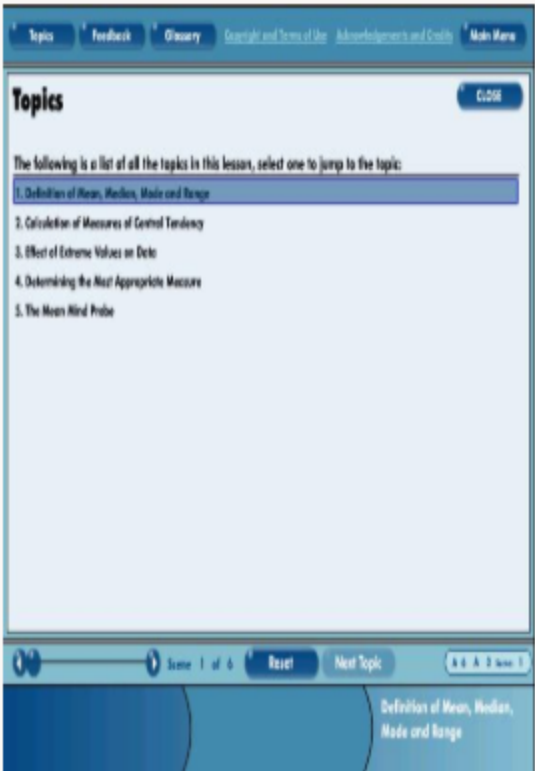
[Central Tendencies](#)
[Save to My Workspace](#)

Description:	This interactive mathematics lesson explores the concept of average values. It explains mean, median, mode, and range, as well as the attributes of th...
Grade:	12
Subject:	Mathematics
Media format:	Web
Learning resource type:	Lesson material, Game
Show More Info	

Step Four:

The clip will automatically start playing.

Select the **Topics** Button and complete all **five** topics listed.



LearnAlberta Information

As you complete the first three topics in the Central Tendencies Exercise, record in your notebook the values for mean, mode, and median for each of the following:

- * hockey example (goals)
- * curling example (final scores)
- * Math 14 test example with 5 tests
- * Math 14 text example with 6 tests

You need these values in Assignment 3.

Measures of Variability

While terms of central tendency refer to how clustered (central) a group of numbers is, measures of variability refer to how spread out a set of numbers is. The terms variability, distribution, and dispersion are synonyms and refer to the spread of a data set. The most commonly used measures of variability (or spread) are the range, percentile, interquartile range, variance, and standard deviation.

The least complicated measure of variability is the range. The range is the difference between the highest value in a data set and the lowest value in a data set.

In the weather example, the range is

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This is the difference between 19 °C (the highest value) and 10 °C (the lowest value).

To understand interquartile range one must first understand percentiles. A percentile is not the same as a percentage. If Lance scored 32 of a possible 50 marks on a test for introversion, his percentage score would be 64%. To determine the percentage, you divide 32 by 50 then multiply the quotient by 100. If Lance wanted to compare his score to the scores of others on the same test, he would use his percentile. If 70 percent of the other scores were less than Lance's score (less than 32 out of 50) then Lance's score would be the 70th percentile. To summarize, percentiles compare your score to the scores of others who have taken the same test.

In any set of data, the total number of percentiles is 100. Researchers may divide the total number of percentiles into 10 equal parts called deciles, or they can divide the total number of percentiles into four equal parts called quartiles. If a person's score was in the first decile, he or she would have scored less than 90% of others on the same test. If a person scored in the fourth quartile, he or she scored higher than 75% of the population who took the test. The median of all percentiles is the 50th percentile. The difference between the first quartile (the 25th percentile) and the third quartile (the 75th percentile) is the interquartile range. This range contains the middle 50% of the scores in a distribution.

range: the difference between the highest value and the lowest value in a data set

percentile: a value on a scale of 100 that indicates the percent of a distribution that is equal to or below it

decile: any one of nine numbers that divide a distribution into 10 parts such that each part contains the same number of scores

quartile: any of the three values that divide a distribution into four parts with each part containing one fourth of the total population

Practice with Percentiles

A group of high school Biology students were asked to identify correctly 14 different kinds of perennial flowers. Below is a chart of scores showing how well each student did on the test.

Sam	Mindy	Wes	Ted	Toni	Fred	Sally	Stacey	Ron	Ann
11/14	2/14	3/14	7/14	5/14	5/14	12/14	6/14	3/14	8/14

Q: What is Ted's *percent*?

A: Divide 7 by 14 then multiply the quotient by 100.

Ted's percent is 50%

Q: What is Ted's *percentile*?

A: To begin, reorder the scores from lowest to highest.

Mindy	Wes	Ron	Toni	Fred	Stacey	Ann	Ted	Sam	Sally
2/14	3/14	3/14	5/14	5/14	6/14	6/14	7/14	11/14	12/14

Ted ranks in eighth position (from the lowest mark) in the class of 10. Because the number of percentiles in any given data set is 100, we must multiply 10 (the number of scores) by another factor to equal 100. In this case, we multiply 10 by 10 to get 100. This means we simply multiply Ted's rank of 8 by 10 to get his percentile.

Ted's percentile is 80.

Q: What is Sam's percentile?

A: Sam's percentile is 90.

Sam's percentile means that 10% of people scored higher on the test than he did and 90% of people scored lower on the test than he did.

In some studies, researchers want to know how close each individual score is to the middle of the distribution. Using the mean as the measure of the middle of a distribution, researchers calculate the

average squared difference of the scores from the mean. The result of this calculation is called the **variance**. The variance is used to determine the standard deviation. The **standard deviation** is the square root of the variance.

Wow... that sounds complicated!

Don't worry, you won't be required to perform these types of calculations in the final exam (or project). However, be familiar with the terms mean, mode, median, range, variance, and standard deviation.

A simple example of how to calculate variance and standard deviation involves a dart game. Six friends each throw one dart at a dart board much like the one in the photo above. Three of the six friends get a "bulls-eye" (dead centre) worth 10 points. Of the other three friends, one lands his dart in the first ring closest the centre and earns 9 points. The second friend lands her dart in the third ring closest the centre and earns 7 points. The last friend lands his dart in the fifth ring closest to the centre and earns 5 points.

Q: What is the standard deviation in the dart game?

Step One: Determine the arithmetic average or **mean** from the data set by adding all six values and dividing the sum by six.

	Score
Person One	10
Person Two	10
Person Three	10
Person Four	9
Person Five	7
Person Six	5
Total	$10 + 10 + 10 + 9 + 7 + 5 = 51$
Average or Mean	$51/6 = 8.5$

Step Two: Calculate the difference between each score and the mean value, and then square that difference.

	Score	Difference from Mean	Square of Difference
Person One	10	$10 - 8.5 = 1.5$	$1.5^2 = 2.25$
Person Two	10	$10 - 8.5 = 1.5$	$1.5^2 = 2.25$
Person Three	10	$10 - 8.5 = 1.5$	$1.5^2 = 2.25$
Person Four	9	$9 - 8.5 = 0.5$	$0.5^2 = 0.25$
Person Five	7	$7 - 8.5 = -1.0$	$-1.0^2 = 1.00$
Person Six	5	$5 - 8.5 = -3.5$	$-3.5^2 = 12.25$

Step Three: Determine the arithmetic average or **mean** of all the squared differences by adding all six calculated values together and dividing the sum by six. This mean is also known as the **variance**.

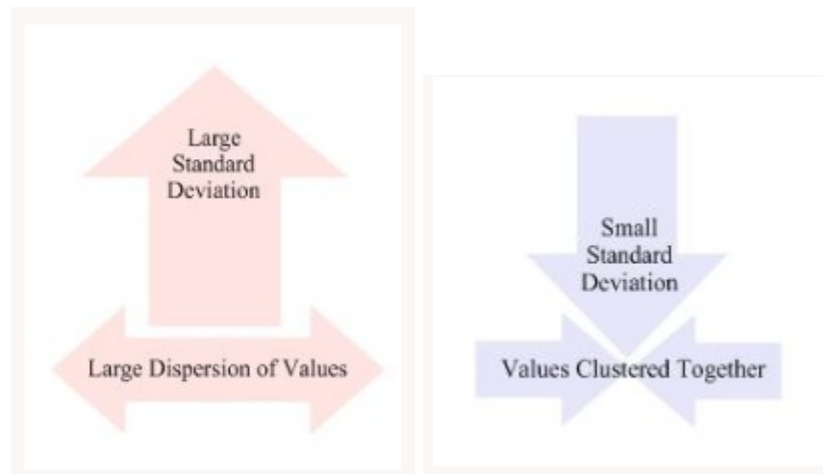
	Square of Difference
Person One	2.25
Person Two	2.25
Person Three	2.25
Person Four	0.25
Person Five	1.00
Person Six	12.25
Total	$2.25 + 2.25 + 2.25 + 0.25 + 1.00 + 12.25 = 20.25$
Average or Mean (Variance)	$20.25 / 6 = 3.375$

Step Four: Determine the standard deviation by finding the square root of the variance.

$$\sqrt{3.375}$$

A: 1.837

If the value of standard deviation is small, the numbers in a data set are clustered closely together. If a standard deviation is large, then the numbers are more spread out. Graphically, this can be represented as follows:



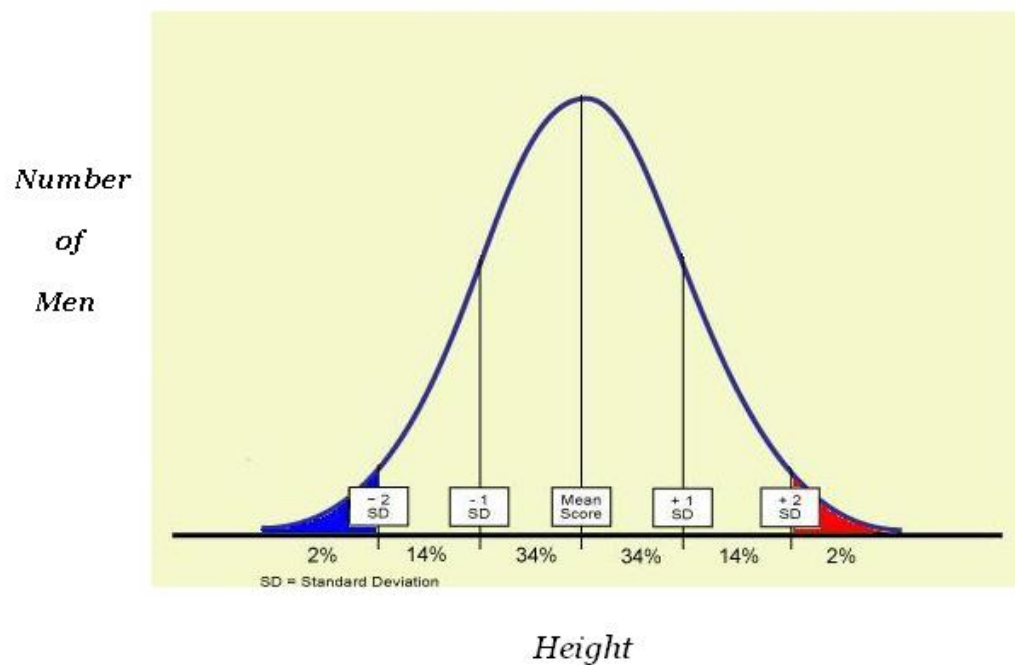
Statistics in Psychology

As you know, all psychologists study behaviour. Many psychologists want to determine what is "normal" behaviour for a given population. In psychology, a simple definition of a **statistical deviation** (from *normal*) could be stated as *a noticeable departure from those traits or behaviours possessed by the majority of people*.

If we use adult height as an example, a noticeable departure from the average or norm would be a seven-foot tall individual (2.13 metres) or a four-foot tall individual (1.22 metres). Because the heights of the majority of adults are less extreme than either of these two examples, we can say that the two heights (seven feet and four feet) are abnormal.

To restate this idea, individuals with characteristics and behaviours that are not within an average range (characteristics falling beyond a particular distance from the average value) can sometimes be considered abnormal. To continue with our height example, we can use 5 feet 10 inches as the mathematical average (mean) height of Canadian men. Many men are either taller than 5 feet 10 inches or shorter than 5 feet 10 inches but are not considered abnormal with respect to height. Scientists can statistically show "abnormal" by collecting data, calculating the average, determining the standard deviation, and then finding all individuals that are farther than two standard deviations away from the average. The individuals farther than two standard deviations from the average generally make up about 5% of the sample population. This can be graphically displayed in a *distribution curve* also known as a *bell curve* or a *normal distribution*.

Bell Curve of Height in a Population of Canadian Men



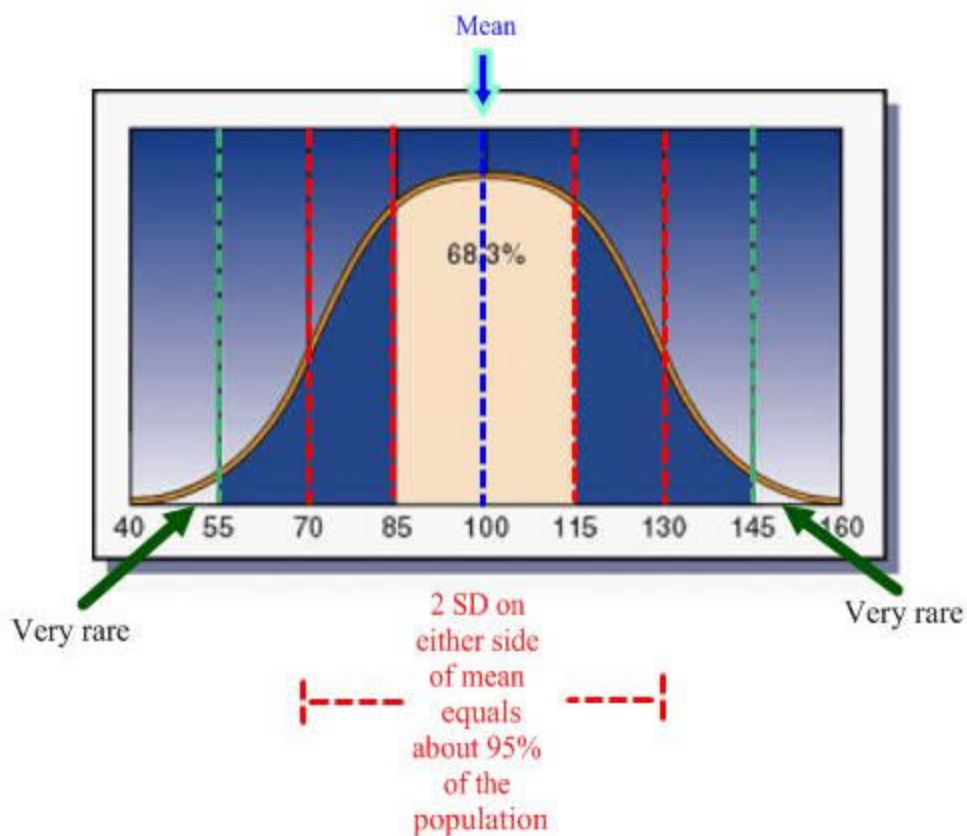
As shown in the figure above, the x-axis represents height and the y-axis represents the number of men sampled. The **blue** area on the left and the **red** area on the right represent those men who are either abnormally short or abnormally tall. These individuals fall more than two standard deviations from the average height or mean score. The area under the graph between the red and blue sections represents those men who are of "normal" height.

statistical deviation: a significant departure from those traits or behaviours possessed by the majority of people

Albertans and IQ

If the entire population of Alberta completed an IQ test, the scores would fall into a bell-shaped curve around the mean of 100. More than two-thirds of the population would score between 85 and 115. About 5% of the population would score less than 70 or more than 130. IQ scores below 55 and above 145 are very rare.

On the diagram below, the x-axis reflects IQ scores and the y-axis is the number of people who completed the test.



Although the statistical method of determining abnormality (characteristics or behaviour falling outside the "norm") is scientific and valid, it does nothing to address certain important issues such as health and welfare. For example, in the past two decades the bell curve for weight (mass) in North America has shifted right (with mass on the x-axis). This means that we, as a society, are on average becoming heavier. This may be the norm, but is it healthy? Is it a good thing?

Other Important Terms and Concepts

Scientists must have a common understanding or definition of scientific concepts and processes so that communication is effective and misunderstandings are reduced. For example, scientists from many countries know that on the periodic table the symbol **K** represents the metal potassium although they may not have the same word associated with the symbol. Below are the flags of five countries and the words they use to describe the element represented by the letter **K**. Can you correctly identify each nationality? Mouse-over the flags for the to see if you guessed correctly.

 Potassium	 Kalium	 Potassio	 Potassium	 Kalij
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Having a common language and common understanding of important concepts is vital to accurate communication. A chemist in the Ukraine might not understand what *l'eau*, *Wasser*, and *el agua* mean (the French, German, and Spanish words for "water"), but he or she would know that $H_2O_{(l)}$ represents that wonderful liquid!

Interesting Fact: Many elements on the periodic table have Latin names. The Latin name for potassium is *kalium* and that is why that element has the symbol K!

For more information about the periodic table, go to <http://www.periodni.com/en/>

Periodic table

Online calculators

Articles and tables

Downloads

English

°C

K

1100

1400

1000

1300

900

1200

800

1100

700

1000

600

900

500

800

400

700

300

600

200

500

100

400

0

300

200

100

0

International Temperature Scale

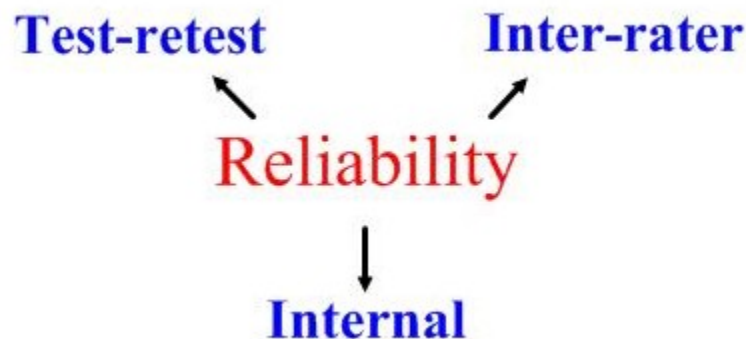
PERIODIC TABLE OF THE ELEMENTS

1																	18
1 H 1.008																	2 He 4.0026
3 Li 6.94	4 Be 9.0122															10 Ne 20.180	
11 Na 22.990	12 Mg 24.305	13 Al 26.982	14 Si 28.085	15 P 30.974	16 S 32.06	17 Cl 35.45	18 Ar 39.948										
19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.38	31 Ga 69.723	32 Ge 72.630	33 As 74.922	34 Se 78.971	35 Br 79.904	36 Kr 83.798
37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.95	43 Tc [98]	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57-71 La-Lu	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
87 Fr [223]	88 Ra [226]	89-103 Ac-Lr	104 Rf [261]	105 Db [270]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [282]	112 Cn [285]	113 Nh [286]	114 Fl [289]	115 Mc [289]	116 Lv [293]	117 Ts [294]	118 Og [294]
Copyright © 1998-2019 by Ems Generalist																	
Lanthanide		57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05	71 Lu 174.97	
Actinide		89 Ac [227]	90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np [237]	94 Pu [244]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]	
Solid		Liquid		Gas		Triple point of water (273.16 K, 0.01 °C)											

Reliability and Validity

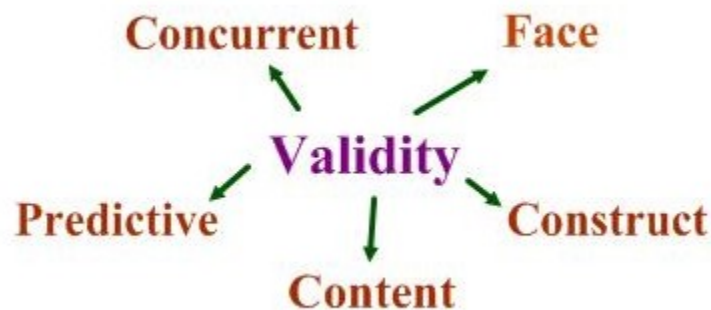
Just as rock climbers expect their ropes and carabiners will reliably bear their masses as they climb upwards and rappel down, researchers in all sciences, not just psychology, want their experiments to be reliable so that others may repeat their experiments and obtain similar results. When experiments are reliable and valid, conclusions reached can be transferred to similar situations in other parts of the world.

Specifically, **reliability** is the extent to which an experiment, test, or measuring procedure yields the same results on repeated trials. Reliability can be divided into three categories: test-retest, internal, and inter-rater. The first category, **test-retest reliability**, requires that the psychologist obtain similar results with the same test or measurement at two different points in time. With **internal reliability**, various parts of a test or measurement must produce similar results. In the last category, **inter-rater reliability**, two or more psychologists (or "raters"), must come to similar conclusions regarding a measurement, test, or experiment.



Experiments, tests, and measurements must also be valid. **Validity** refers to the degree to which a technique or test measures what it is designed to measure. Validity can be divided into five categories, the first of which is face validity. As the category implies, **face validity** refers to a test or technique appearing to measure what it is supposed to measure. With **content validity**, the test or technique used should actually measure the content or areas that are most important. The test or technique used may not measure everything, and it does not have to -- as long as it measures the major areas of concern.

The third category, **concurrent validity** requires that similar tests or experiments yield the same results. For example, two different I.Q. tests taken by the same individual should yield similar results. As the name suggests, **predictive validity** refers to how well behaviour can be predicted. For example, different patients with the same diagnosis are predicted to respond the same way to the same treatment. The last category of validity, **construct validity**, requires that a test measures what it is designed to measure. For example, if a test is designed to measure musical ability, but instead it measures verbal reasoning, it cannot be considered a valid test.



reliability: the extent to which an experiment, test, or measuring procedure yields the same results on repeated trials

test-retest reliability: similar results are obtained from the same test at two different points in time

internal reliability: different parts of a test produce similar results

inter-rater reliability: two or more psychologists reach similar conclusions using the same test

validity: a test measures what it is designed to measure

face validity: a test appearing to measure what it is supposed to measure

content validity: a test measuring the most important content

concurrent validity: similar tests yield similar results

predictive validity: the ability to predict behaviour based on test results

construct validity: a test measures what it is designed to measure

You have reached the end of the content for
Section One of Module Two.